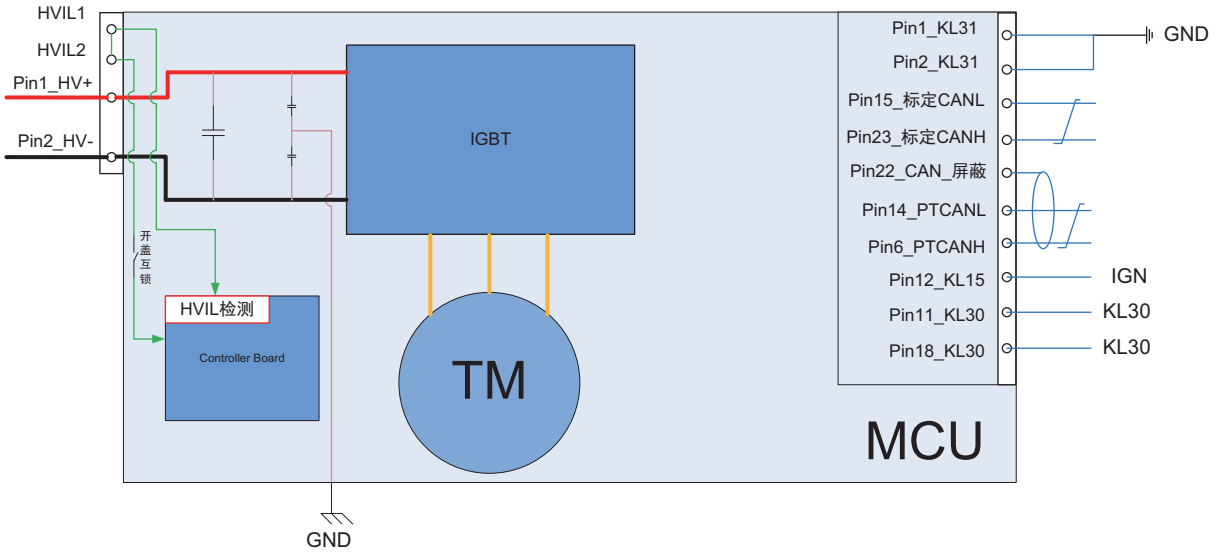
P0A44-37、 P1A20-37

Fault diagnosis code
P0A44-37：Motor overspeed
P1A20-37：High motor speed alarm
Detection tools
1 Multimeter and diagnostic apparatus
Possible causes
• Circuit,terminal and adapting piece fault • PEU fault • High voltage insulation fault • Bus voltage detection circuit in the PEU fault
Conditions for fault code operation
Turn the ignition switch to ON position.
Conditions for fault code setting
P0A44-37
The absolute speed of the motor is greater than 17000rpm
P1A20-37
The absolute value of the motor speed is greater than 16300rpm to report the fault, than 16200rpm begins to reduce than power, and the speed from 16600rpm to 0.
To set the effect of a fault code condition
When vehicle lost power, there is forced power at high speed

Circuit diagram



EV901001

Troubleshooting steps	
Step 1	General inspection.
	1 Check the fuse for damage. 2 Check battery capacity. 3 Check the harness plug of PEU for damage, poor contact, aging and looseness. 4 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	
Step 2	Check fault codes.
	1 Connect the fault diagnosis instrument to the diagnosis interface. 2 Turn the ignition switch to ON position. 3 Diagnose the GAW module with diagnostic apparatus. 4 If there is any fault code except for P0A44-37,P1A20-37?
	YES → Refer to:DTC Summary list(PEU).
NO ↓	
Step 3	Check BMS.
	1 Turn the ignition switch to OFF. 2 Connect the fault diagnosis instrument to the diagnosis interface. 3 Turn the ignition switch to ON position. 4 Diagnose the BMS with diagnostic apparatus. 5 Test whether the faulted code is existed.
	YES → DTC diagnosis is performed based on fault codes.
NO ↓	
Step 4	Check the connecting cable of PEU.
	1 Turn the ignition switch to OFF. 2 Make protective preparations according to the safety operation specifications of the high voltage voltage in the maintenance manual. 3 Disconnect the harness plug battery pack high voltage. 4 Disconnect the high voltage harness plug of PEU. 5 Check whether the PEU connection cable is rotten, damaged, and water enters. 6 Check whether the result is normal or not?

	NO → Repair or replace the faulted parts.
YES ↓	
Step 5	Check the insulation of PEU.
	1 Use a megohm meter to check the insulation resistance of the PEU housing and grounding according to the procedure in the maintenance manual. **Standard value: >20MΩ** 2 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	
Step 6	Check the insulation of driving motor.
	1 Use a megohm meter to check the insulation resistance of the driving motor housing and grounding according to the procedure in the maintenance manual. **Standard value: >20MΩ** 2 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	
Step 7	Check the high voltage relay.
	1 Turn the ignition switch to OFF. 2 Make protective preparations according to the safety operation specifications of the high voltage voltage in the maintenance manual. 3 Open the PEU end cover. 4 Check whether the high voltage relay works. 5 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	
Step 8	Perform initialization.
	1 Turn the ignition switch to OFF. 2 Connect the fault diagnosis instrument to the diagnosis interface. 3 Turn the ignition switch to ON. 4 The governor was initialized with the diagnostic instrument. 5 Can the governor be initialized and is the governor restored?
	NO → Repair or change the governor.
YES ↓	

Step 9	Check PEU internal circuit.	
		1 Turn the ignition switch to OFF. 2 Make protective preparations according to the safety operation specifications of the high voltage voltage in the maintenance manual. 3 Open the PEU end cover. 4 Check the PEU internal circuit and components for obvious damage or . ablation. 5 Check whether the result is normal or not?
		NO → Repair or replace the faulted parts.
YES ↓		
Step 10	Test whether PEU is norm.	
		1 Turn the ignition switch to OFF. 2 Disconnect and check the harness plug EB1002 of PEU. 3 Check and repair. • Corrosion(install new adapting piece or terminal-cleaning module and pin). • Pin damaged or bent-install new terminal/pin. 4 Fly out of pins-install new pins as needed. 5 Reconnect the PEU plug and all other previously disconnected plug. Make sure the pin is properly positioned and fixed. 6 Run the system and determine if the problem persists?
		YES → Change the PEU Module
NO ↓		
At this point the system is functioning normally. The problem may be caused by module connections. Address the root cause of any connector or pin problems.		
Next Step ↓		
Diagnosis end.		